

Project Documentation

Large Scale Performance Lab using Python

Objective

Develop a Python application to simulate multiple users accessing a website simultaneously, measure response times, and generate a basic performance report.

Project Scope

The project demonstrates basic performance testing concepts using multithreading. Multiple HTTP requests are sent to a target website, response times are measured, and the results are summarized in a report.

Tools & Technologies

Python 3.x, Requests Library, Threading Module, Visual Studio Code, Git, GitHub.

Project Structure

Source code/performance_lab.py
README.md
Requirements.txt
Output/performance_report.txt
Screenshots/
Documentation/

Working Principle

1. Set the target URL.
2. Create multiple threads to simulate concurrent users.
3. Send HTTP GET requests.
4. Measure response times.
5. Display a performance summary.
6. Save the report to performance_report.txt.

Features

Concurrent user simulation, response time measurement, HTTP status monitoring, average response time calculation, and automatic report generation.

Expected Output

Displays the status code and response time for each simulated user, followed by a summary containing the total users, successful requests, and average response time. A performance_report.txt file is generated.

Future Enhancements

Add configurable user counts, CSV/HTML reports, graphical charts, response time percentiles, and integration with professional load-testing frameworks.

Conclusion

This project introduces the fundamentals of large-scale performance testing using Python and demonstrates concurrent request execution, response analysis, and automated report generation.